

ESSEY ARAYA

859-528-8771 | araya.essey@gmail.com | [linkedin.com/in/esseyaraya/](https://www.linkedin.com/in/esseyaraya/) | essey.netlify.app | github.com/essey1

EDUCATION

Berea College

Berea, KY

Bachelor of Science in Engineering Physics and Computer Science, GPA: **3.93/4.00**

Expected Dec 2027

- Honors & Awards: 4-year tuition promise scholarship (\$285,000+ USD), Dean's List (All Semesters)
- Relevant Coursework: Machine learning, Numerical Analysis, Linear Algebra, Differential Equations, Classical Mechanics, Data Structures & Algorithms, Embedded Systems, Design & Analysis of Algorithms

EXPERIENCE

Machine Learning Engineer Intern

Sep 2026 – Present

NASA × Break Through Tech

Remote / New York, NY

- Developing a RAG and agentic AI system to automate software requirements auditing against NASA engineering standards, reducing compliance review time from weeks to minutes.

Machine Learning Research Intern

May 2026 – Aug 2026

Vanderbilt School of Engineering

Nashville, TN

- Built and integrated GP-PHS into NeuroMANCEER using Python, PyTorch, and GPyTorch, enabling researchers to model dynamical systems with uncertainty-aware machine learning
- Designed and implemented 9 modular components for data pipelines, matrix operations, and numerical computation using NumPy, Pandas, and SciPy, providing reusable infrastructure for ML training
- Learned a 4-state nonlinear dynamical system with $< 5\%$ error, generalizing to 3 test trajectories
- Held weekly Agile review meetings with a 6-member team, incorporating feedback to iteratively develop comprehensive API documentation and 5 tutorial notebooks
- Created 250+ unit, integration, and end-to-end tests using Pytest to improve reliability

Mechanical Engineering Intern

May 2025 – Jul 2025

Berea College

Berea, KY

- Designed and fabricated a lightweight composite battery enclosure for Berea College's first solar car using Ansys ACP composite simulator and wet layup, reducing projected manufacturing cost by 50%
- Eliminated 80% of testing costs by designing composite layups and conducting FEA in Ansys Mechanical

PROJECTS & PUBLICATIONS

Gaussian Process Port-Hamiltonian Systems | Submitted to IFAC SYSID 2027

Sep 2026

Lumen - Team Lead | Next.js 16, TypeScript, PostgreSQL, Vercel AI SDK

[View Project](#)

- Built a greedy prerequisite-graph solver in TypeScript to automate 4-year scheduling for 1,500+ students
- Developed a course-ranking pipeline that translated students' career goals into 3 personalized degree pathways, helping students align coursework with their desired careers

SafeSource - Hackathon | Python, FastAPI, Gemini Vision API, Pydantic v2, Next.js 15

[View Project](#)

- Built a platform that analyzed environmental data to help users identify health risks by quantifying PFAS
- Developed a Google Gemini Vision + Pydantic pipeline to estimate PFAS exposure from grocery receipt images, helping users identify high-risk products in everyday purchases

Drone Vertical Dynamics Simulation | C++, Python, GitHub Actions (CI/CD)

[View Project](#)

- Built a C++ physics simulator modeling gravity, thrust, and drag, producing velocity and height data
- Implemented CI/CD with GitHub Actions to compile, run, validate, and deploy outputs automatically

Biscute | HTML, CSS, SCSS, JavaScript

[View Project](#)

- Built a website for a local small business expanding customer reach and increasing revenue by \$5,000

LEADERSHIP

President

Mar 2026 – Present

Berea Engineering and Racing

Berea, KY

- Led a 3-member team to optimize the chassis, reducing weight by 32% while meeting safety targets
- Led weekly meetings and organized hackathons and design competitions, reaching 100+ participants

TECHNICAL SKILLS

Languages: C++, Python, MATLAB, TypeScript, JavaScript, SQL

ML & Developer Tools: PyTorch, NumPy, SciPy, Pandas, Matplotlib, Git, Linux, Pytest, GitHub Actions